

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 8 - Summative Assessment 2

Summative Assessment (SA-2) - Answer Key

Class: Class 8

Assessment Type: Summative Assessment (SA-2)

Total Marks: 40

Duration: 2 hours

Topics Covered: Algorithms, Emerging Technologies, Cyber Safety, Computational Thinking

Answer Key - Grading Guidelines

Note to Teacher: This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

Part A: MCQ (10 x 1 = 10 marks)

Q.No	Answer	Explanation
1	c	Merge Sort has $O(n \log n)$ average-case time complexity
2	c	Binary search has $O(\log n)$ time complexity on sorted arrays
3	b	Ambiguity is NOT a characteristic of a good algorithm
4	b	IoT stands for Internet of Things
5	c	Blockchain is the foundation of cryptocurrency
6	b	Flowcharts represent algorithms visually
7	b	On-demand self-service is a characteristic of cloud computing
8	b	Diamond symbol represents decision/condition in flowcharts
9	b	Machine learning is a subset of AI that enables systems to learn from data
10	c	Enabling two-factor authentication is most important cyber safety practice

Part B: Short Answer (8 x 2 = 16 marks)

1. Write algorithm to find largest element and analyse time complexity.

• **Answer:**

Algorithm: FindLargest

1. Set max = first element
2. For each element i from 2 to n :
 - a. If element $i > \text{max}$, set max = element i
3. Return max

Time complexity: $O(n)$ - linear time as we traverse the list once.

- **Marking:** 1 mark for correct algorithm, 1 mark for time complexity analysis.

2. Differentiate between AI, ML, and DL.

- **Answer:** AI is broad field of creating intelligent machines. ML is subset of AI that learns from data without explicit programming. DL is subset of ML using neural networks with multiple layers. Examples: AI - Chess game, ML - Email spam filter, DL - Image recognition.
- **Marking:** 1 mark for explaining each concept, 1 mark for examples.

3. Explain divide and conquer with Merge Sort example.

- **Answer:** Divide and conquer breaks problem into smaller subproblems, solves them recursively, combines solutions. Merge Sort divides array in half recursively until single elements, then merges sorted halves. Example: $[38, 27, 43, 3] \rightarrow [38, 27]$ and $[43, 3] \rightarrow$ further division and merging.
- **Marking:** 1 mark for explaining divide and conquer, 1 mark for Merge Sort example.

4. What is Big Data? Explain 5 V's.

- **Answer:** Big Data refers to extremely large datasets that cannot be processed by traditional tools. 5 V's: Volume (massive data size), Velocity (speed of data generation), Variety (different data types), Veracity (data accuracy and quality), Value (useful insights from data).
- **Marking:** 1 mark for Big Data definition, 1 mark for explaining 5 V's.

5. **Write Python program for linear and binary search.**

- **Answer:**

```
def linear_search(arr, target):
    for i in range(len(arr)):
        if arr[i] == target:
            return i
    return -1
```

```
def binary_search(arr, target):
    low, high = 0, len(arr) - 1
    while low <= high:
        mid = (low + high) // 2
        if arr[mid] == target:
            return mid
        elif arr[mid] < target:
            low = mid + 1
        else:
            high = mid - 1
    return -1
```

Comparison: Linear $O(n)$, Binary $O(\log n)$ but requires sorted array

- **Marking:** 1 mark for linear search, 1 mark for binary search.

6. **Explain three cyber safety measures.**

- **Answer:** 1) Use strong, unique passwords - prevents unauthorized access. 2) Enable two-factor authentication - adds extra security layer. 3) Don't click suspicious links - protects against phishing and malware. Each measure protects different aspects of online safety.
- **Marking:** 1 mark for each measure with explanation.

7. **What is cloud computing? Differentiate IaaS, PaaS, SaaS.**

- **Answer:** Cloud computing delivers computing services over internet. IaaS: Infrastructure as Service (e.g., AWS EC2 - virtual machines). PaaS: Platform as Service (e.g., Google App Engine - development platform). SaaS: Software as Service (e.g., Google Docs - ready-to-use software).
- **Marking:** 1 mark for cloud computing definition, 1 mark for differentiating IaaS, PaaS, SaaS.

8. **Write pseudocode for bubble sort and trace on [5, 3, 8, 1, 2].**

- **Answer:**

```
BubbleSort(arr):
    For i = 0 to n-1:
        For j = 0 to n-i-2:
            If arr[j] > arr[j+1]:
                Swap arr[j] and arr[j+1]
```

Trace: Pass 1: [3,5,1,2,8], Pass 2: [3,1,2,5,8], Pass 3: [1,2,3,5,8], Pass 4: [1,2,3,5,8]

- **Marking:** 1 mark for pseudocode, 1 mark for trace.

Part C: Long Answer (4 x 3 = 12 marks)

1. **Write Python program for Insertion Sort with trace and analysis.**

- **Answer:** Complete program with insertion sort implementation, trace on [12, 11, 13, 5, 6], and analysis: Best case $O(n)$ when sorted, Average/Worst case $O(n^2)$ when reverse sorted.
 - **Marking:** 1 mark for correct program, 1 mark for trace, 1 mark for complexity analysis.
2. **Explain Blockchain, IoT, and AI with examples.**
- **Answer:** Blockchain: Distributed ledger technology, blocks linked cryptographically, application in supply chain management. IoT: Connected devices exchanging data, example in smart traffic lights. AI: Machines mimicking human intelligence, applications in personalized learning and medical diagnosis.
 - **Marking:** 1 mark for each technology with example.
3. **Design traffic management system using algorithms.**
- **Answer:** Problem statement, algorithm for signal timing based on vehicle density, flowchart for decision process, IoT sensor integration for real-time data collection and adaptive signal control.
 - **Marking:** 1 mark for problem statement and algorithm, 1 mark for flowchart, 1 mark for IoT integration.
4. **Design library system with algorithms, SQL, and cloud computing.**
- **Answer:** Search algorithm with error handling, SQL database design for books/members/loans, Python pseudocode for search function, cloud computing for remote access and data backup.
 - **Marking:** 1 mark for algorithm, 1 mark for SQL design, 1 mark for cloud integration.

Part D: Application Based (2 x 1 = 2 marks)

1. **Explain how technology helps social media company.**
- **Answer:** Big Data analytics processes user behavior patterns to understand preferences. Machine Learning algorithms analyze viewing history to recommend content. AI uses natural language processing to detect and remove harmful content automatically.
 - **Marking:** 1 mark for Big Data and ML explanation, 1 mark for AI explanation.
2. **Explain how technology helps farmer.**
- **Answer:** IoT sensors monitor soil moisture, temperature, and humidity in real-time. Data analytics processes historical data to predict optimal planting and harvesting times. Cloud computing provides access to weather forecasts and market prices from anywhere.
 - **Marking:** 1 mark for IoT and data analytics, 1 mark for cloud computing explanation.